

Exp 6

Write a program to implement Naïve Bayes algorithm in python and to display the results using confusion matrix and accuracy.

Code :

```
import numpy as np

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import confusion_matrix, accuracy_score
```

```
X = np.array([
    [2, 60],
    [3, 65],
    [5, 75],
    [6, 80],
    [8, 90],
    [1, 55],
    [7, 85],
    [4, 70]
])
```

```
y = np.array([
    "Fail",
    "Fail",
    "Pass",
    "Pass",
    "Pass",
    "Fail",
    "Pass",
    "Fail"
])
```

```
model = GaussianNB()
```

```
model.fit(X, y)
```

```
y_pred = model.predict(X)
```

```
cm = confusion_matrix(y, y_pred)
```

```
acc = accuracy_score(y, y_pred)
```

```
print("Confusion Matrix:")
```

```
print(cm)
```

```
print("\nAccuracy:", acc * 100)
```

```
test = np.array([[7, 82]])
```

```
prediction = model.predict(test)
```

```
print("\nTest Sample:", test)
```

```
print("Predicted Class:", prediction[0])
```

```
output:
```

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/yuvas/Downloads/win-g2010-1_3-n_mcd/win/LIB/machine learning lab/exp 6.py
Confusion Matrix:
[[4 0]
 [0 4]]

Accuracy: 100.0

Test Sample: [[ 7 82]]
Predicted Class: Pass
>>> |
```